

Qualcomm's Chris Patrick on Snapdragon 8 Gen 2, Indian innovation and battery tech

What does Qualcomm think about increased competition and innovation?

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Launched in November 2022, Qualcomm's Snapdragon Gen 2 chip is going to find itself in a lot of premium flagship smartphones of 2023, promising cutting-edge performance enhancements in almost all critical departments of a smartphone's functioning. Qualcomm isn't hiding the fact that the Snapdragon Gen 2 has quite a few AI innovations baked into the chip itself, something that the average user wouldn't even notice every day, for instance.

"How we train AI and use it in the chip really depends heavily on the context, because we use AI for a lot of different things," says Chris Patrick, senior vice president and general manager of mobile handsets, Qualcomm Technologies, Inc.

"For example, we use AI a lot for image processing. For different types of different types of visual effects as part of the camera, as part of the still images and video capture and processing. Yeah, so we have a lot of domain specific knowledge and domain specific expertise as part of several other kinds of camera-specific adaptive algorithms," Patrick explains.

He further highlights how Qualcomm has been working on these algorithms for many years, training these AI algorithms with lots of data to best optimise what Qualcomm thinks the best end user preference will be. But that's far from all where Patrick says Qualcomm deploys AI-on-chip



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on the Snapdragon Gen 2. "We have a lot of AI deployed in the phone's modem. That's one of the things we announced, the Snapdragon X70 5G modem, earlier in the year. It uses AI for things like beam steering, which is part of millimetre wave (mmWave) processing or channel state estimation for better 5G throughput, for example. So these are some of the problem areas that Qualcomm has been studying for over 30 years, where we use AI to help process vast amounts of data available in modem-level signal processing and try to make the optimal decision for the real-time, dynamic operation of the device. AI gives us a chance to bring different kinds of tools to solve all the prob-

lems we've been studying over the years. That's not all that Chris Patrick discussed with us during his recent visit to India in December 2022. While speaking from Bengaluru, Patrick underscored the importance of critical research and development conducted out of Qualcomm India, revealing a small insight into the role the India team had to play in Snapdragon Gen 2's innovation journey.

"I've been at Qualcomm for 26 years, shouldering many roles on Qualcomm's engineering side, which kept bringing me to India over the years. Most of the projects I've been a part of over the past decade and a half, our India team has had a strong role to play in its delivery and execution," Patrick emphasises. he underscores.

"The Qualcomm India team is absolutely essential, really indispensable in terms of everything we do all over the world. On my current trip, one of the things we'll be talking about is the strong role on some of the advanced camera applications. Some of the cutting edge end-to-end camera capabilities we demonstrated in Snapdragon Gen 2 were really birthed out of innovations and execution done by the Qualcomm India team," Patrick reveals, explaining further how some of the advanced multimedia applications Qualcomm develops and deploys across devices, the India team plays a critical role on both the system engineering side as well as hardware implementation side and then the final software implementation.

We asked Qualcomm's Chris Patrick if modern smartphones are plateauing in terms of disruptive innovations, if they're in danger of becoming mundane commodities, and if the disruptive tech will emerge out of wholly